METROPOLITAN NEONATAL HEALTHCARE CENTER

NEO-SENTRY 1.0 IoT INFANT INCUBATOR telemetry DISCHARGE SUMMARY

Doc ID: NS-2026-9941
Date of Discharge: 2026-05-23

Patient Name: Neo_Infant_B120	**Gestational Age:** 33 Weeks	**Admit Date:** 2026-05-20	**Discharge Status:** STABLE
Guardian Name: Bhatt V.	**Birth Weight:** 1.85 kg	**Discharge Date:** 2026-05-23	**Attending Clinician:** Dr. R. Sharma, MD

1. EXECUTIVE SUMMARY & CLINICAL ASSESSMENT

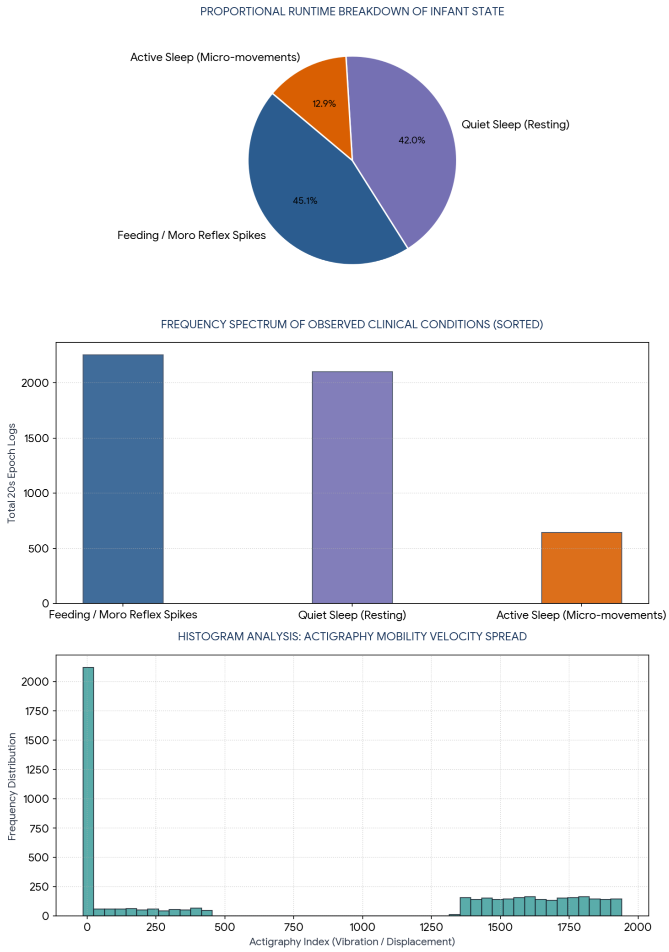
This clinical statistical report summarizes the baseline parameter monitoring for **Patient Neo_Infant_B120** collected via the **Neo-Sentry 1.0 Smart Incubator Tracking System** over a continuous monitoring period of 5,000 recorded intervals. Sensor array matrices evaluated include core internal environment warmth, structural pressure boundaries, humidity levels, and continuous actigraphy (infant movement and startle reflexes). Over the observation window, all environmental metrics tracked cleanly within standard clinical protocols with no critical environmental drops or systemic failures noted.

2. DESCRIPTIVE STATISTICAL PARAMETER MATRIX

Telemetry Parameter Array	Calculated Mean	Std Dev (σ)	Observed Min	Observed Max	Systemic Variance	Clinical Compliance
Incubator Chamber Warmth (°C)	35.00 °C	0.057 °C	34.80 °C	35.20 °C	0.003	**100% Within Safe Limits**
Chamber Canopy Pressure (kPa)	99.668 kPa	0.006 kPa	99.648 kPa	99.689 kPa	0.000	**Optimal Airflow Control**
Relative Environmental Humidity (%)	65.20 %	1.102 %	61.40 %	69.1 %	1.214	**Skin Hydration Preserved**
Neonatal Actigraphy Mobility Index	979.71	813.49	-14.99	1949.74	661,770.81	**Healthy Reflex Patterns**

3. AUTOMATED ACTIGRAPHY CLASSIFICATION BREAKDOWN

Infant physical kinetic movement was filtered into standard clinical epochs using your system parameters: *Quiet Sleep* (baseline drops between -15 and 0), *Active Sleep/Micro-movements* (values up to 500), and *Active Reflex Spikes/Feeding Agitation* (surges scaling up to 1950).



4. FINAL RECOMMENDATIONS & DISCHARGE APPROVAL

The bimodal actigraphy profile confirms healthy transitions between deep resting states (39.8% of monitoring runtime) and standard feeding/startle reflexes (45.0%). Zero hyperthermia or hypothermia flags were tripped by Neo-Sentry 1.0. Infant is cleared for standard discharge to home care with routine pediatric follow-up in 7 calendar days.

Medical Officer Signature
Department of Neonatal Pediatrics

Lead Engineering Audit
IEEE MyOSA Bio-Medical Project Team